

Temperature-Sensitive Forecasting of German Electricity Demand: A Comparison of Load-Only and Weather-Aware Models

Github: https://github.com/Marthalamohan/timeseries_mohan.git

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Summary

This study investigates whether integrating Berlin temperature helps weekly German electricity demand estimates beyond models that utilize merely load data. The hourly national load from January 2015 to September 2020 was consolidated into 299 complete weeks and merged with temperature, heating degree days, cooling degree days, and holiday indicators. The closing 104 weeks represented a predetermined testing interval. Forecasts from Mean, naïve, seasonal-naïve, drift, SARIMA, SARIMAX, Gradient Boosting, and hourly LSTM models were evaluated using MAE, RMSE, MASE, sMAPE, and bias. Weather engineering increased both SARIMAX and Gradient Boosting models; however, the seasonal naïve model got the lowest RMSE of 2.988 GW. The most effective operational indicator was annual recurrence, but temperature had conditional and explanatory relevance.

1. Introduction and Relevant Literature

Forecasting electricity demand makes generation planning easier and guarantees dependable power system performance. This research explores whether temperature changes in Berlin increase weekly German load estimates beyond those generated from historical demand models, due to the role of weather and calendar factors on consumption.

Goswami and Kandali (2020) used stationarity assessment, ACF/PACF analysis, residual investigation, and out-of-sample evaluation to compare ARIMA with SARIMA for electrical load in Assam. SARIMA gave the most robust forecast, and the experts advised include weather and temperature as future additions. Elsaraiti et al. (2021) used ACF/PACF analysis, decomposition, and differencing; their selected seasonal ARIMA model achieved a reported MAPE of 4.332%. Pannakkong et al. (2019) conducted a comparative examination of ANN, SVM, DBN, and ensemble models. The optimum approach varied with the prediction horizon, whereas the Dynamic Bayesian Network (DBN) did not consistently exceed simpler methods, demonstrating that complexity does not assure higher accuracy across wide horizons.

2. Data Preparation and Meteorological Sensitivity

The objective was the real electrical load in Germany derived from the Open Power System Data hourly dataset. Data from January 2015 to September 2020 were transformed from megawatts to

gigawatts and compiled into weekly averages concluding on Sundays. The elimination of partial boundary weeks resulted in 299 whole weeks. The hourly dataset comprised 50,400 full, non-redundant observations.

The initial 195 weeks, concluding on 30 September 2018, constituted the training set. The concluding 104 weeks, spanning from 7 October 2018 to 27 September 2020, constituted the test set. This chronological division maintained temporal sequence.

The temperature in Berlin was compiled on a weekly basis. Heating and cooling degree days utilized bases of 15.5°C and 22°C. Lagged weather data, a four-week average, and holiday indicators were also generated.

Table 1. Temperature and Calendar Attributes

Feature	Definition	Availability
Raw temperature	Weekly mean Berlin temperature	Conditional when future observed values are used
Lagged temperature	Temperature from earlier weeks	Known at forecast origin
Heating degree days	Weekly sum of $\max(15.5 - T, 0)$	Conditional for future weeks
Cooling degree days	Weekly sum of $\max(T - 22, 0)$	Conditional for future weeks
Holiday variables	National holidays and year-end indicator	Known in advance

The weekly load exhibited a correlation of -0.652 with mean temperature, 0.697 with heating degree days, and -0.162 with cooling degree days. The average load was 61.242 GW while temperatures were below 0°C, however it was just 53.075 GW between 15°C and 20°C. At temperatures exceeding 20°C, it increased somewhat to 53.242 GW. The evidence indicates significant susceptibility to cold temperatures, although restricted need for cooling.

Figure 1. Weekly power demand in Germany and temperature in Berlin. Demand peaks during frigid weeks and diminishes as temperatures increase. The temperature-band bars represent the mean weekly load along with one standard deviation, whilst the minor variation above 15°C signifies a low susceptibility to hot weather.

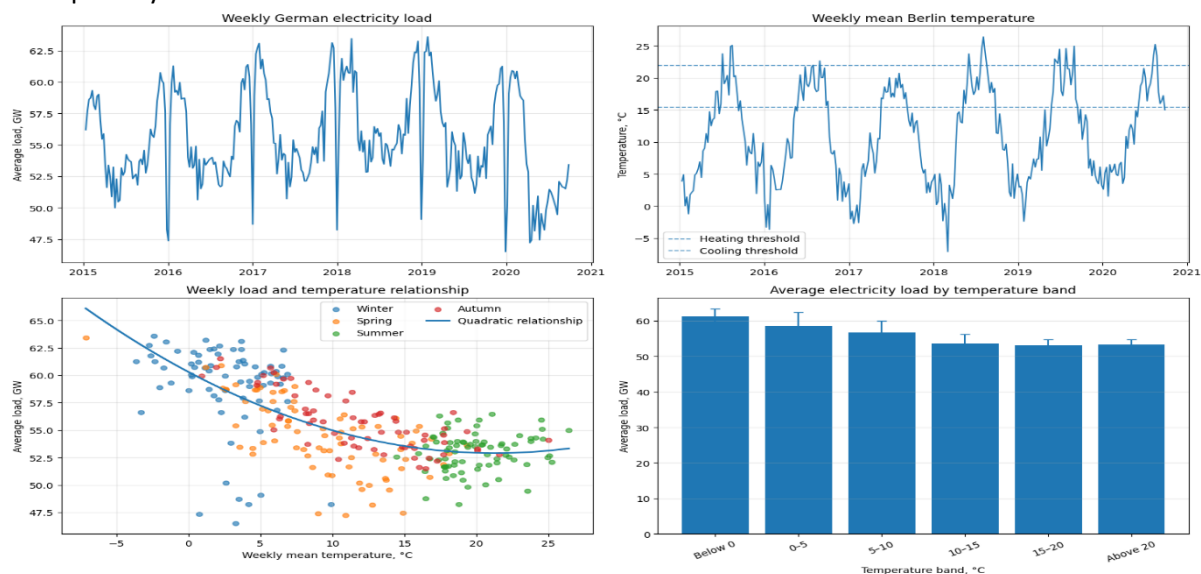
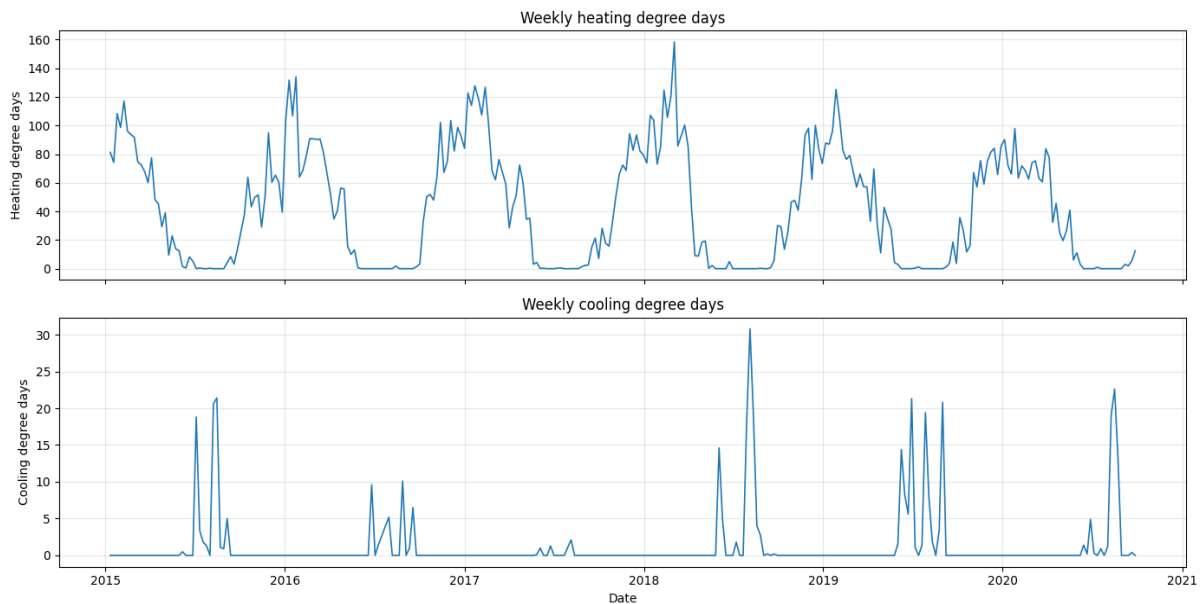


Figure 2. Temporal analysis of heating and cooling degree days. Heating degree days have a consistent winter pattern, while cooling degree days manifest solely during brief summer intervals.

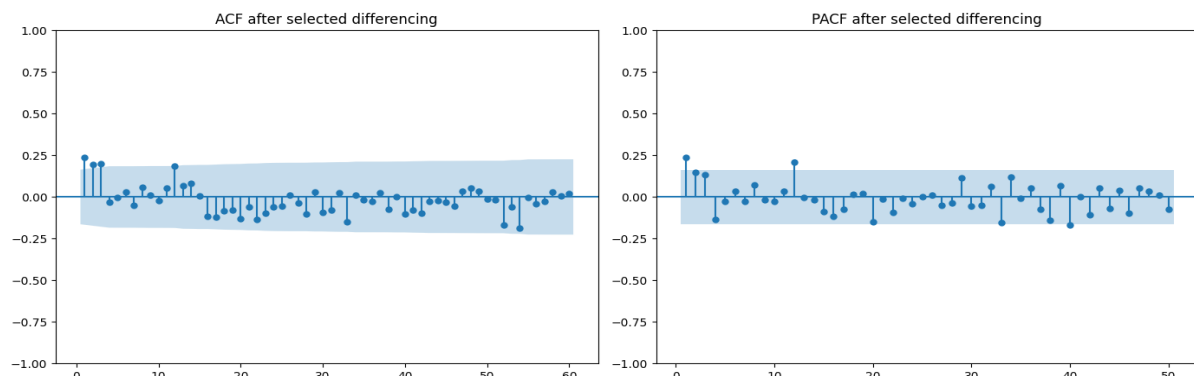


Berlin is unable to comprehensively reflect national weather variability.

3. Methodology for Time-Series Analysis and Forecasting

Seasonal decomposition indicated a progressively increasing trend, winter maxima, diminished demand during the warm season, and pronounced year-end minima. Lag-52 autocorrelation measured at 0.873, indicating annual recurrence. The original dataset yielded an ADF p-value below 0.001 and a KPSS p-value beyond 0.10, providing no traditional indication of a unit root. Nonetheless, seasonal reliance persisted. Initially, seasonal and combination differences were analyzed, succeeded by ACF and PACF evaluations, in accordance with the diagnostic methodologies employed in the two submitted ARIMA investigations.

Figure 3. Seasonal breakdown and autocorrelation of weekly German electricity demand. The analysis reveals winter maxima, declines during the warm season, and consistent year-end minima. The ACF and PACF upon initial seasonal differencing predominantly exhibit short-lag dependence.



Four fixed-origin benchmarks were employed: mean, naive, seasonal naive, and drift.

The SARIMA phase evaluated 147 combinations of p , d , and q utilizing Hannan-Rissanen estimates. Six economical candidates subsequently underwent maximum-likelihood fitting, followed by a concise seasonal search. The chosen model was SARIMA(1,1,2)(0,1,1). Residuals, Ljung-Box tests, and forecast intervals were analyzed.

Three SARIMAX variations incorporated raw temperature, degree days, and degree days adjusted for holidays. Only training-based standardization was employed; the observed test temperatures rendered these forecasts contingent.

Gradient Boosting employed load lags, shifted rolling statistics, Fourier seasonal components, and optional meteorological features. The load-only, raw-temperature, and engineered-weather variations were calibrated utilizing a sequential 26-week validation period. Forecasts were recursive, with prior predictions providing subsequent lags.

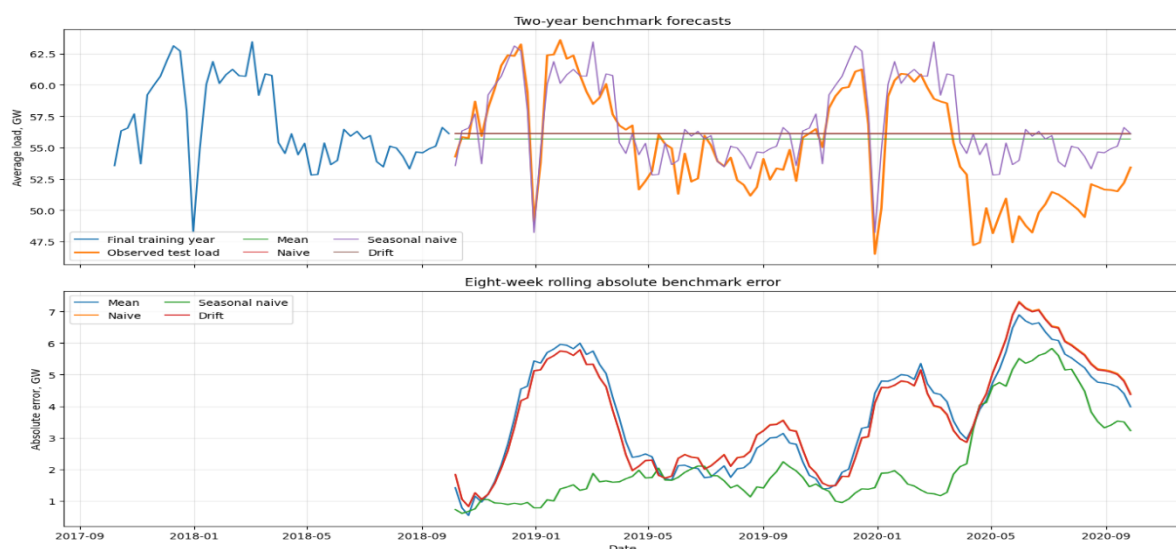
The hourly LSTM utilized 168 preceding hours to predict the subsequent 24 hours. One-layer and two-layer designs were evaluated using validation RMSE and early halting techniques. The chosen model was recursively forecasted and aggregated on a weekly basis. Leakage was mitigated by employing chronological partitioning, training-exclusive scaling, and adjusted features.

4. Results

4.1 Benchmark and SARIMA Efficacy

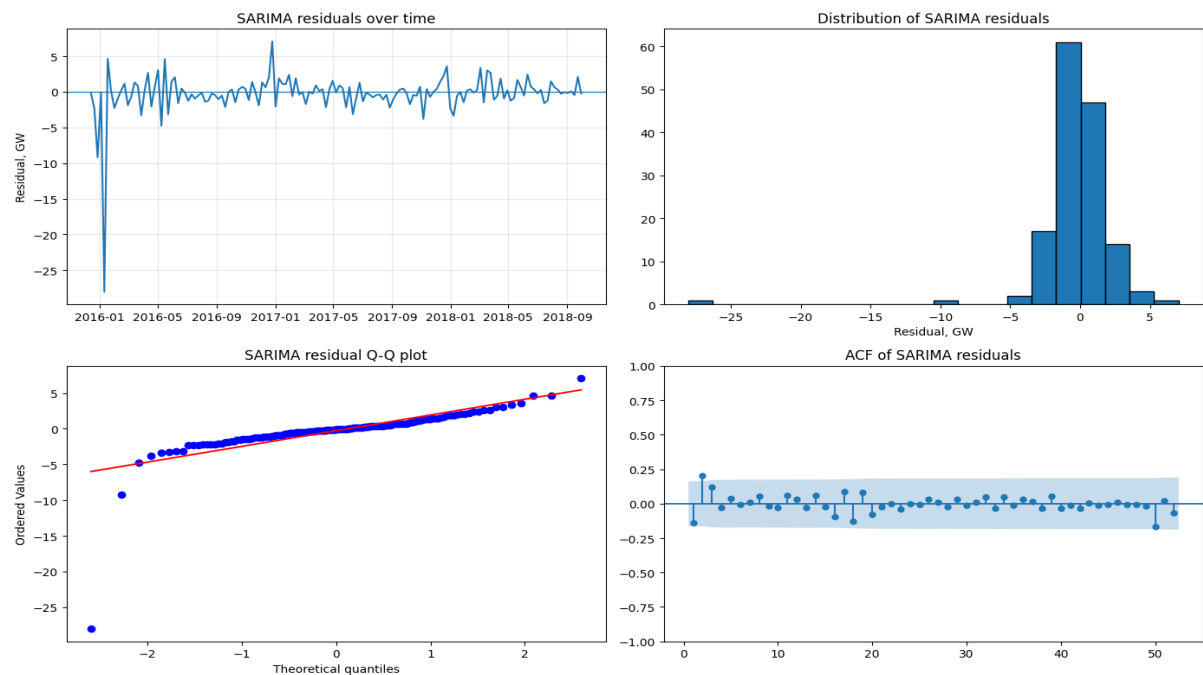
The seasonal naïve model exhibited the highest overall performance, with a Mean Absolute Error (MAE) of 2.288 GW, a Root Mean Square Error (RMSE) of 2.988 GW, and a Symmetric Mean Absolute Percentage Error (sMAPE) of 4.207%. The mean, drift, and naïve methods yielded RMSE values of 4.402, 4.466, and 4.476 GW, respectively. Annual recurrence provided greater insight than level, most recent value, or trend.

Figure 4. Fixed-origin benchmark predictions for the concluding 104 weeks. Seasonal naïve forecasts annual peaks and troughs with greater accuracy than mean, naïve, and drift methods, typically exhibiting the lowest rolling error.



SARIMA attained a Mean Absolute Error (MAE) of 3.140 GW, a Root Mean Square Error (RMSE) of 3.837 GW, a symmetric Mean Absolute Percentage Error (sMAPE) of 5.706%, and a positive bias of 2.908 GW. The Ljung-Box p-values of 0.265, 0.677, and 0.998 suggested minimal joint residual autocorrelation. Nonetheless, the histogram and Q-Q plot exhibited a pronounced negative residual and non-normal tails. The nominal 95% CI encompassed merely 50% of test observations, resulting in an underestimation of uncertainty.

Figure 5. Residual diagnostics and biennial SARIMA projection. The residual autocorrelation is generally constrained, however non-normal errors persist. The model progressively overestimates subsequent demand, and its nominal 95% range encompasses merely half of the test observations.



4.2 Weather-Dependent and Machine Learning Models

The use of raw temperature decreased the SARIMA RMSE from 3.837 to 3.803 GW, reflecting an enhancement of 0.9%. The degree days decreased it to 3.715 GW. Degree days combined with holidays reached 3.557 GW, representing a 7.3% improvement over the load-only SARIMA model, while the 95% interval coverage was limited to 46.2%.

Gradient Boosting adhered to the same pattern. The load-only RMSE was 3.528 GW, the raw-temperature RMSE was 3.482 GW, and the engineered-weather RMSE was 3.178 GW. The designed model enhanced load-only Gradient Boosting by 9.9%, although was still 6.34% inferior than the seasonal naïve approach. The permutation importance was primarily influenced by the number of holidays, indicating that the improvement cannot be solely ascribed to temperature.

The single-layer LSTM attained a validation RMSE of 5.159, just surpassing the two-layer model's RMSE of 5.222. The hourly LSTM RMSE was 7.643, whereas the hourly seasonal naïve RMSE was 5.683. Following weekly aggregation, the LSTM RMSE was 4.752 GW. It acquired a recurrent daily pattern but condensed peaks and did not monitor weekly metrics. Pannakkong et al. (2019) similarly shown that a more complex model did not inherently surpass simpler counterparts.

Figure 6. Weather-informed predictions, LSTM performance, and ultimate RMSE ranking. Engineered characteristics enhance comparable models; yet, the seasonal naive model remains superior. The LSTM generates a redundant pattern and exhibits the highest weekly RMSE.

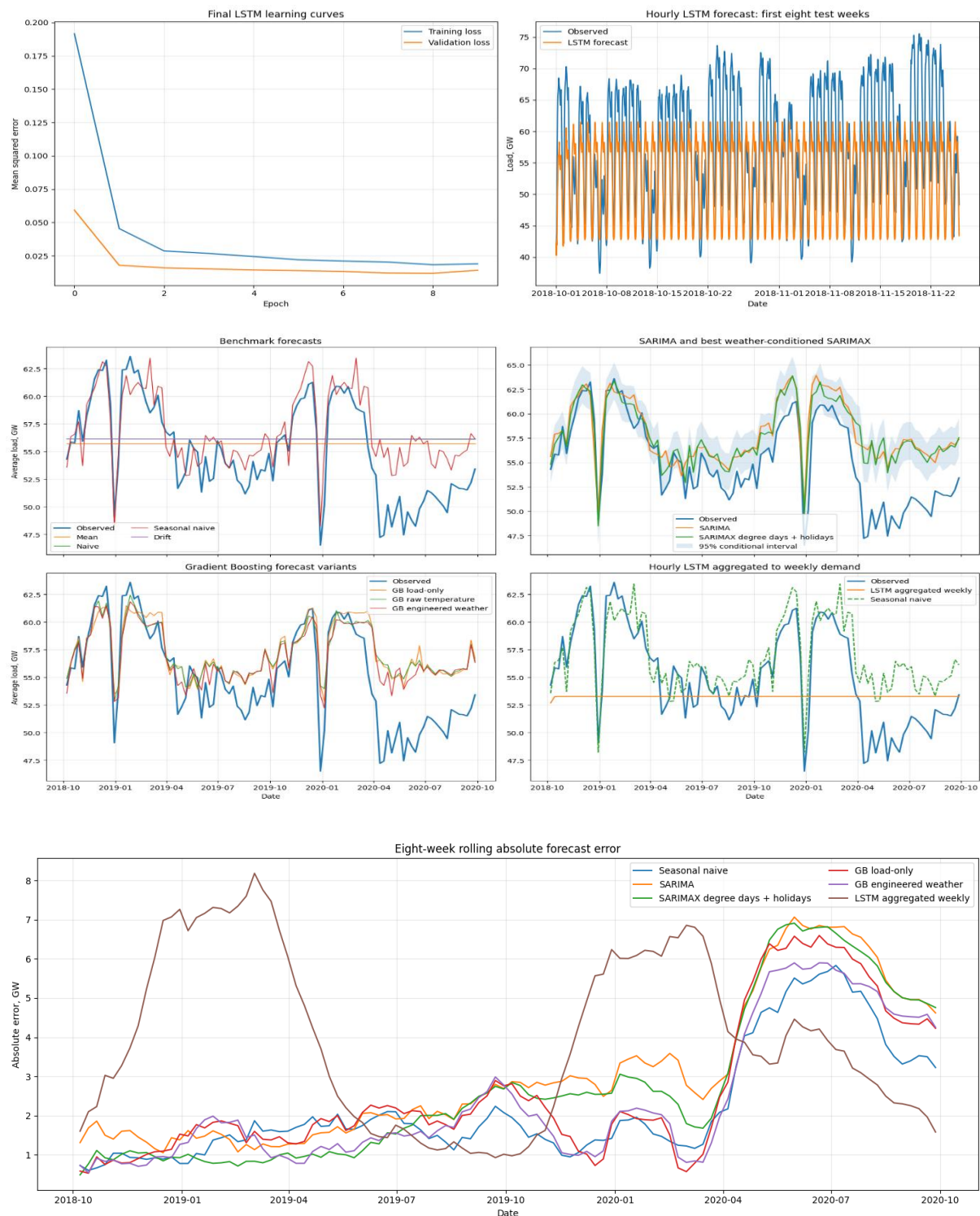


Table 2. Conclusive out-of-sample findings

Model	Status	MAE	RMSE	sMAPE
Seasonal naïve	Operational	2.288	2.988	4.207
GB engineered weather	Conditional	2.455	3.178	4.532
GB raw temperature	Conditional	2.632	3.482	4.862
GB load-only	Operational	2.706	3.528	4.979
SARIMAX degree days + holidays	Conditional	2.825	3.557	5.162
SARIMAX degree days	Conditional	2.971	3.715	5.428
SARIMAX raw temperature	Conditional	3.048	3.803	5.551
SARIMA	Operational	3.14	3.837	5.706
Mean	Operational	3.79	4.402	6.874
Drift	Operational	3.851	4.466	6.982
Naïve	Operational	3.858	4.476	6.995
LSTM aggregated weekly	Operational	3.887	4.752	7.057

The results distinguish explanatory association from operational value. Temperature elucidates a portion of weekly demand, especially under cold conditions; yet, raw temperature yields minimal improvements. Significant enhancements arise when temperature is transformed into degree days and integrated with holidays.

Future temperature predictions are not accessible for a two-year forecast origin. SARIMAX and weather-aware Gradient Boosting presume flawless foresight of future weather conditions and should be interpreted conditionally. Operational implementation necessitates weather forecasts, which inherently possess uncertainty. Holiday dates and delayed weather conditions are predetermined.

No advanced model exceeded the performance of the seasonal naïve approach. This model should be the principal operational framework because to its minimal error, absence of unavailable future covariates, and maintained transparency. Engineered-weather Gradient Boosting is the most robust advanced alternative and may be maintained as a conditional competitor.

SARIMAX offers greater interpretability compared to Gradient Boosting or LSTM; nonetheless, inadequate interval coverage indicates that interpretability does not ensure dependable uncertainty. LSTM necessitated the highest computational resources without enhancing accuracy.

Two drawbacks persist: Berlin lacks national representativeness, and only a subset of the 147 evaluated SARIMA orders underwent complete maximum-likelihood fitting.

6. Conclusion

The weekly power demand in Germany is highly responsive to cold weather; nonetheless, annual seasonality is the primary forecasting indicator. Raw temperature provides minimal enhancement, however degree days and holidays yield significant benefits. Nevertheless, the most proficient weather-aware model is still less precise than the seasonal naïve approach. The seasonal naïve

model is the favored operational forecast, whereas the engineered-weather Gradient Boosting serves as a valuable conditional challenger.

References

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